

THE UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY



Kiểm thử phần mềm - 22KTPM3

DATA GENERATION - HW3

Thành Viên

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Chapter 1

Group Information

Group ID: 07

Member Name	Student ID	Assigned Tables	Status
Cao Uyển Nhi	22127310	- Invoices	Done
		- Invoice Items	Done
Lưu Thanh Thuý	22127410	- Brands	Done
		- Categories	Done
Nguyễn Phước Minh Trí	22127424	- Product Image	Done
		- Personal Access Token	Done
Võ Lê Việt Tú	22127435	- Product	Done
		- Favorites	Done
Trần Thị Cát Tường	22127444	- User	Done
		- Contact Requests	Done

Chapter 2

Data Generation

2.1 Table Selection

- I selected the tables **personal_access_tokens** and **product_images**. The `personal_access_tokens` table is essential for user authentication and security, while the `product_images` table supports product display, directly impacting customer experience and revenue.

2.2 List of Data Fields and Rules

- **Table** `personal_access_tokens`:

Field	Type	Rule
<code>id</code>	Integer	Auto-incremented from 1 to 500.
<code>tokenable_id</code>	Integer	Random integer from 1 to 100.
<code>tokenable_type</code>	String	Randomly selected from “App\User”, “App\Admin”.
<code>name</code>	String	“Token #1” to “Token #500”.
<code>token</code>	String	Random 64-character string (generated using UUID).
<code>abilities</code>	String	Randomly selected from [“read”, “write”, “read,write”].
<code>last_used_at</code>	Timestamp or NULL	Randomly NULL or between 2024-01-01 and 2025-06-24.
<code>expires_at</code>	Timestamp or NULL	Randomly NULL or between 2024-01-01 and 2025-06-24.
<code>created_at</code>	Timestamp	Between 2024-01-01 and 2025-06-24.
<code>updated_at</code>	Timestamp	Between <code>created_at</code> and 2025-06-24.

- **Table** product_images:

Field	Type	Rule
id	Integer	Auto-incremented from 1 to 500.
product_id	Integer	Random integer from 1 to 200.
name	String	Randomly selected from ["hammer_image.jpg", "saw_image.jpg", "drill_image.jpg"].
by_user	String	Randomly selected from ["admin", "user1", "user2"].
source	String	Randomly selected from ["website", "upload", "external"].
file_name	String	"prod_001_hammer.jpg" to "prod_500_drill.jpg".
title	String	Randomly selected from ["Hammer Image", "Saw Image", "Drill Image", etc ...].
created_at	Timestamp	Between 2024-01-01 and 2025-06-24.
updated_at	Timestamp	Between created_at and 2025-06-24.

- **Screenshots:**

```

test2.py > ...
1  from faker import Faker
2  import pandas as pd
3  import random
4
5  fake = Faker()
6  data = {
7      'id': range(1, 501),
8      'product_id': [random.randint(1, 200) for _ in range(500)],
9      'name': [f"{random.choice(['hammer', 'saw', 'drill'])}_image.jpg" for _ in range(500)],
10     'by_user': [random.choice(["admin", "user1", "user2"]) for _ in range(500)],
11     'source': [random.choice(["website", "upload", "external"]) for _ in range(500)],
12     'file_name': [f"prod_{i:03d}_{random.choice(['hammer', 'saw', 'drill'])}.jpg" for i in range(1, 501)],
13     'title': [f"Hình {random.choice(['búa', 'cưa', 'khoan'])}" for _ in range(500)],
14     'created_at': [fake.date_time_between(start_date="-1y", end_date="now") for _ in range(500)],
15     'updated_at': [fake.date_time_between(start_date="-1y", end_date="now") for _ in range(500)]
16 }
17 df_images = pd.DataFrame(data)
18 df_images.to_excel('product_images.xlsx', index=False)

```

Figure 2.1: Table personal_access_tokens

- **Explanation of Process and Steps**

I used Python with the faker library to generate data for the two tables. The process included:

1. Installing the faker and pandas libraries (using pip install faker pandas).
2. Writing Python code to generate random data according to the defined rules.
3. Exporting the data to two Excel files: personal_access_tokens.xlsx and product_images.xlsx, each containing 500 rows.
4. Manually verifying the data to ensure it was meaningful (e.g., no duplicate tokens, valid timestamps).

```

test2.py > ...
1  from faker import Faker
2  import pandas as pd
3  import random
4
5  fake = Faker()
6  data = {
7      'id': range(1, 501),
8      'product_id': [random.randint(1, 200) for _ in range(500)],
9      'name': [f"{random.choice(['hammer', 'saw', 'drill'])}_image.jpg" for _ in range(500)],
10     'by_user': [random.choice(['admin', "user1", "user2"]) for _ in range(500)],
11     'source': [random.choice(["website", "upload", "external"]) for _ in range(500)],
12     'file_name': [f"prod_{i:03d}_{random.choice(['hammer', 'saw', 'drill'])}.jpg" for i in range(1, 501)],
13     'title': [f"Hình {random.choice(['búa', 'cưa', 'khoan'])}" for _ in range(500)],
14     'created_at': [fake.date_time_between(start_date="-1y", end_date="now") for _ in range(500)],
15     'updated_at': [fake.date_time_between(start_date="-1y", end_date="now") for _ in range(500)]
16 }
17 df_images = pd.DataFrame(data)
18 df_images.to_excel('product_images.xlsx', index=False)

```

Figure 2.2: Table product_images

- **Explanation of Source Code:**

I developed a custom Python script to automate data generation. For example, for the personal_access_tokens table:

1. Used range(1, 501) to generate id.
2. random.randint(1, 100) to generate tokenable_id.
3. fake.uuid4()[64] to create a random token string.
4. fake.date_time_between() to generate random timestamps. The data was stored in a pandas DataFrame and exported to Excel.

- **Sample Data:**

id	tokenable_id	tokenable_type	name	token	abilities	last_used_at	expires_at	created_at	updated_at
1	67	App\Admin	Token #1	1a879790-1c7a-4a4e-b67c-b75f27b14742	read		2026-01-21 00:47:41	2024-07-10 20:05:10	2024-12-12 17:32:26
2	55	App\User	Token #2	8cff5fbd-ebf7-428d-8055-db1a4014be88	write			2024-12-15 03:41:51	2024-07-11 00:07:20
3	53	App\User	Token #3	4d148a2f-acb0-4d4a-9929-03c905200dad	read	2025-03-02 15:50:24	2026-03-13 14:59:57	2025-06-16 20:02:44	2025-01-30 05:21:11
4	99	App\Admin	Token #4	a89bc542-c3cc-43db-b7a1-ff02f2e2e733	write		2026-06-18 19:51:40	2024-10-14 07:34:32	2024-10-12 05:56:39
5	15	App\Admin	Token #5	80018fa6-0d21-4c9a-8d4c-7b7155dab0da	write	2024-08-15 06:03:28		2025-05-09 05:01:44	2024-10-27 17:03:57
6	56	App\User	Token #6	7f62fb5c-dd37-465f-bfed-213ae3267743	write	2024-08-05 20:20:32	2025-07-05 15:29:43	2024-09-09 09:33:31	2024-09-13 12:46:31
7	62	App\User	Token #7	a61d914c-c96a-442a-b351-d3454d08d1ad	read		2026-05-10 11:25:44	2025-02-28 16:20:08	2025-01-05 20:56:44
8	92	App\Admin	Token #8	f141772d-cfc1-4175-a632-f37972684942	read	2024-07-22 11:09:16	2025-07-15 07:20:16	2024-10-14 14:07:17	2024-08-17 18:40:46
9	30	App\Admin	Token #9	801ce011-f286-4629-bbd7-fadb7eaf7d21	write	2025-06-07 17:41:34		2025-04-13 19:15:20	2024-12-06 14:45:18
10	12	App\User	Token #10	27047c83-de54-4401-9936-f3796e1f0007	read		2026-03-28 20:48:50	2024-11-27 20:53:14	2024-10-19 17:49:52

Figure 2.3: 10 sample rows personal_access_tokens

id	product_id	name	by_user	source	file_name	title	created_at	updated_at
1	114	hammer_image.jpg	user2	external	prod_001_saw.jpg	Hình cưa	2024-07-13 06:40:26	2024-12-05 01:07:10
2	13	saw_image.jpg	user1	external	prod_002_hammer.jpg	Hình cưa	2025-05-24 18:43:11	2024-11-19 10:35:51
3	130	hammer_image.jpg	admin	external	prod_003_hammer.jpg	Hình cưa	2025-03-20 00:01:16	2025-01-23 01:51:09
4	187	hammer_image.jpg	user1	upload	prod_004_drill.jpg	Hình khoan	2025-02-25 19:04:48	2025-04-05 09:06:15
5	41	drill_image.jpg	user2	external	prod_005_drill.jpg	Hình búa	2024-08-04 13:54:58	2024-10-04 08:06:33
6	130	saw_image.jpg	admin	external	prod_006_drill.jpg	Hình cưa	2024-07-01 19:58:53	2024-11-08 15:12:30
7	17	drill_image.jpg	admin	website	prod_007_saw.jpg	Hình khoan	2024-11-08 15:03:59	2024-10-29 00:06:30
8	26	drill_image.jpg	user1	external	prod_008_saw.jpg	Hình búa	2025-04-08 13:53:28	2025-03-07 21:54:52
9	199	saw_image.jpg	user1	upload	prod_009_hammer.jpg	Hình cưa	2024-07-17 04:16:11	2025-03-02 03:46:20
10	113	hammer_image.jpg	admin	upload	prod_010_saw.jpg	Hình cưa	2025-05-01 04:27:30	2024-11-10 00:00:14

Figure 2.4: 10 sample rows product_images

Chapter 3

Self-Evaluation (Data Generation)

Criteria	Description	Max Point	Self-assessment
2 Tables Selection	2 important tables selected	1.0	1.0
Sample data	All data must be meaningful	2.0	2.0
Data generation report	Report is clear, traceable, professional, with self-assessment	1.0	1.0